Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	5	**1**	
1(b)	$\frac{3}{20}$ cao	**1**	
1(c)	$\frac{9}{20}$ oe	**2**	**M1** for $1-\frac{11}{20}$ or $20-11$ or $3+6$ oe
1(d)(i)	Mode	**1**	
1(d)(ii)	Biology	**1**	
1(d)(iii)	Correct reason	**1**	
1(e)(i)	11 198 3 54 6 108	**3**	**B2** for 2 correct angles OR **B1** for 11, 3, 6 **M1** for $\frac{k}{20}\times 360$ seen, $k=1,11,3,6$ or one of *their* frequencies
1(e)(ii)	Correct pie chart drawn	**2**	**B1FT** for one correct sector drawn, provided sum of *their* 3 angles is 360
2(a)	87.5	**1**	
2(b)	–2.5	**2**	**M1** for $21-26.7$ or $26.7-3.2$
2(c)	431.9[0]	**2**	**M1** for $47.6+70.2+16.7$ oe or $2\times 128+2\times 20.7$ oe or **B1** for 134.5, 148.7, 297.4, 58.1 or 373.8 or 283.2

Question	Answer	Marks	Partial Marks
2(d)	31 800	**3**	**M2** for $3\times\frac{90}{100}\times4000$ oe or $7\times\frac{75}{100}\times4000$ oe or **M1** for $\frac{90}{100}\times4000$ oe or $\frac{75}{100}\times4000$ oe
2(e)	$\frac{684}{0.0129}$	**M1**	or $684-51400\times0.0129$
	$53023-51400$	**M1**	or $\frac{20.94}{0.0129}$
	1623	**A1**	
2(f)	2641.5 2642.5	**2**	**B1** for each If 0 scored, **SC1** for answers correct but reversed
3(a)	72	**1**	
3(b)	Correct reason	**1**	
3(c)(i)	Enlargement [centre] $(-3, 4)$ [scale factor] 3	**3**	**B1** for each
3(c)(ii)	Rotation [centre] (0,0) oe 180° oe	**3**	**B1** for each
3(d)(i)	Correct translation $(3,2),(4,4),(6,2)$	**2**	**B1** for translation $\begin{pmatrix}7\\k\end{pmatrix}$ or $\begin{pmatrix}k\\-1\end{pmatrix}$
3(d)(ii)	Correct reflection $(-1,-5)$, $(-3,-7)$, $(-4,-5)$	**2**	**B1** for reflection in $y=k$, $k\neq-1$ or in $x=-1$
4(a)	*J* marked in correct position	**2**	**M1** for bearing 036 from *K* or bearing 284 drawn from *L*
4(b)(i)	800	**2**	**M1** for $\frac{9600}{k}$ $11.8\leqslant k\leqslant12.2$ soi
4(b)(ii)	12 23	**3**	**M1** for $\frac{9.6}{4.5}$ oe or **B1** for figs 213 or 2133.. **A1** for 2h 8min If **A0** scored, **SC1** for 10 15 + *their* time correctly evaluated